Title: AI-Powered Plagiarism Detection: A Web-Based System for Text Integrity

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Type of conference / Journal :

IEEE international conference paper.

Publisher :

IEEE (Institute of electrical and electronics Engineers)

Abstract :

The paper addresses the growing issue of plagiarism in academic and professional content, especially with the rise of digital text creation tools. Traditional plagiarism detection systems often rely on pre-indexed databases and fail to provide real-time, paragraph-level analysis. To overcome these limitations, the authors propose a web-based AI-powered plagiarism detection system that uses machine learning and semantic similarity techniques.

The system works by dynamically querying web sources and analyzing text at the paragraph level. It computes similarity scores and provides a plagiarism percentage for each paragraph, offering more granular insights than existing tools. The proposed system aims to improve accuracy, scalability, and usability while addressing the shortcomings of conventional plagiarism detection methods.

Introduction :

Plagiarism, defined as presenting someone else's work as one's own, has become a serious concern in academic institutions and professional environments. With the availability of vast online resources and AI-based content generation tools, detecting plagiarism has become increasingly complex. Traditional tools such as Turnitin and Grammarly primarily rely on pre-indexed databases, which limits their ability to detect newly published or paraphrased content.

The paper introduces a novel system that improves plagiarism detection by analyzing text at a paragraph level rather than document level. It integrates real-time web searches and semantic similarity techniques, making it capable of detecting both direct copying and paraphrased content. This approach enhances detection accuracy and provides more meaningful insights to users.

Identified Research Gap :

The authors identify several limitations in existing plagiarism detection systems. Most traditional tools depend heavily on static databases, which restrict their ability to detect plagiarism from newly available online content. Additionally, many systems focus on document-level analysis rather than paragraph-level granularity, which reduces precision in identifying specific plagiarized sections.

Another major gap is the inability of existing systems to effectively detect paraphrased plagiarism. Methods based on keyword matching or basic similarity metrics often fail when the original content is reworded. Moreover, real-time analysis is lacking in many tools, making them less effective in dynamic digital environments. The proposed system addresses these gaps by combining semantic analysis, machine learning, and real-time web querying, ensuring more accurate and comprehensive plagiarism detection.

Methodology :

The proposed system is implemented as a web-based application using Python and Flask. It allows users to upload documents or input text directly. The system processes the input by dividing it into paragraphs for detailed analysis. Each paragraph is then sent as a query to a web search API (such as Google Custom Search) to retrieve relevant content from the internet.

The core of the system uses Sentence-BERT, a deep learning model, to generate vector embeddings for both the input text and retrieved web content. These embeddings are compared using cosine similarity to measure how closely the texts match. If the similarity score exceeds a predefined threshold, the paragraph is flagged as plagiarized.

The methodology also includes preprocessing steps such as tokenization, stop-word removal, and text cleaning. A machine learning classifier is trained using datasets like the PAN plagiarism corpus to distinguish between plagiarized and original content. The system presents results through a color-coded heatmap, where green indicates low plagiarism, yellow indicates moderate plagiarism, and red indicates high plagiarism.

This structured approach ensures accurate detection, efficient processing, and user-friendly output.

Results :

The results demonstrate that the proposed system achieves high accuracy in detecting plagiarism, with approximately 94% detection accuracy, which is comparable to or better than many existing tools.

One of the key outcomes is the system's ability to provide paragraph-level plagiarism scores, enabling users to identify specific sections of concern. The integration of real-time web queries allows the system to detect plagiarism from newly published or online sources, which is a significant improvement over traditional tools.

The system also provides detailed reports, including similarity scores, matched URLs, and classification of plagiarism types such as direct copying, paraphrasing, and structural similarity. The visual representation through heatmaps further enhances user understanding and usability. Overall, the results validate the effectiveness, scalability, and practicality of the proposed system.

Conclusion :

The paper concludes that the proposed AI-powered plagiarism detection system is a significant advancement in ensuring text originality and maintaining academic integrity. By combining machine learning, semantic similarity, and real-time web search, the system overcomes the limitations of traditional plagiarism detection tools.

The ability to analyze text at a paragraph level and provide detailed plagiarism percentages makes the system highly effective for identifying complex plagiarism patterns, including paraphrasing. Its scalability and user-friendly design make it suitable for both academic and professional applications.

In summary, the system not only improves detection accuracy but also provides actionable insights, making it a valuable tool for students, researchers, and professionals. The research lays a strong foundation for future developments in AI-based plagiarism detection systems.